A Computationally Tractable Model for Modeling Key Nuclides in Fluid-Fueled Molten Salt Reactors

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ABSTRACT

Depletion simulations of liquid fueled molten salt reactors commonly do not distinguish between in-core and ex-core regions, and instead will irradiate the entire volume. Various methods have been proposed to account for the flow of the fuel salt between regions, but a major hurdle is the computational cost. This work investigates a spatially resolved method to account for the flow and chemical removal in a rapid and accurate manner. This spatially resolved method is compared with the spatially unresolved scaled flux method to determine the effectiveness of each method compared to higher fidelity results. This work finds that the scaled flux method is sufficient for modeling most nuclides unless there is a large chemical removal rate, in which case a spatially resolved method is needed.

Keywords: Molten Salt Reactor, Depletion, Chemical removal

1. INTRODUCTION

Liquid-fueled molten salt reactor (MSR) development started in the 1950s with the aircraft reactor experiment [1]. Progress then continued in the molten salt reactor experiment later in the 1960s [2]. After that, more focus was placed on solid fueled light water reactors, which have since become the most common nuclear reactor used in the industry. However, at the Generation IV International Forum in the early 2010s, the MSR was designated as one of the six advanced reactors of interest moving forward [3].

Because interest has shifted back to MSRs in the past 20 years, many previous simulation codes and methods must to be altered to account for the moving fuel. In particular, depletion simulations, which are used to model the temporal evolution of the fuel, are of interest. In solid fueled reactors, methods would consider only the evolution of the fuel in time, since the fuel would not experience advection or diffusion. However, MSRs have the fuel circulate through the in-core and ex-core regions, which must be modeled with different neutronic properties and include advection. Modeling this behavior accurately is vital, as the fuel composition affects reactor dynamics, reactor safety, and safeguards.

Several methods have been demonstrated to account for the movement of the fuel. One approach is very computationally inexpensive, and is referred to as the "scaled flux" method [4]. However, this approach does not model the in-core and ex-core regions in a physical manner. Instead, the scaled flux method scales the flux by the fraction of fuel salt in-core compared to ex-core. Another approach is to calculate the steady-state concentration, changing the system of ordinary differential equations to spatial rather than temporal relations [5].

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Other approaches, such as MCNP/ADDER or the method proposed by Tano et al. using Griffin and Pronghorn, try to model the flow more explicitly [6, 7]. This approach is more computationally expensive than the other approaches that only require solving a system of ordinary differential equations. However, these works have shown that including advection in depletion directly affects the equilibrium concentration of xenon-135 by approximately 45% [7, 8].

However, including advection in depletion has been referred to as computationally "intractable" [4, 7]. This is because tracking all nuclides over space and time requires far more computational work than handling traditional time dependent system of ordinary differential equations (ODE). Methods for solving the spatially resolved system of partial differential equations (PDE) require accurately capturing short time scale flow effects, which makes multi-year long depletion simulations exceedingly expensive.

However, Tano et al. showed that not all nuclides must be modeled while retaining accurate results [7]. The present work takes this idea a step further, and isolates isobars of interest. These isobars capture the beta-decay parents, providing higher fidelity than models of a single nuclide. High-fidelity thermal hydraulic simulations are computationally expensive, so instead, a simple advection model is used to capture the in-core and ex-core effects. With the simplifications implemented, this method can run on an laptop in minutes, rather than requiring a supercomputer allocation.

2. METHODS

The one-dimensional PDE form of the equation to be solved for each nuclide, i, in this work is shown in Equation (1). The ODE form which only contains time dependence is given in Equation (2).

$$\frac{\partial N_i(z,t)}{\partial t} + \nu_i(z,t)\frac{\partial N_i(z,t)}{\partial z} = S_i(z,t) - \mu_i(z,t)N_i(z,t) \tag{1}$$

$$\frac{dN_i(t)}{dt} = S_i(t) - \mu_i(t)N_i(t) \tag{2}$$

In these equations, N_i is the atom density in $\frac{atoms}{cm^3}$, t is time in seconds (s), ν_i is the linear flow rate in $\frac{cm}{s}$, z is the linear distance traveled in cm, S_i is the production, or source, term in $\frac{atoms}{cm^3 s}$, and μ_i is the loss term in s^{-1}. Diffusion is neglected because molten salt reactors are an advection-dominated system [9].

The source term is comprised of decay sources and fission yields in this work. Transmutation is not included, as it would require inclusion of many more nuclides, which would increase the computational cost beyond what is desired for this work. The loss term includes losses from both decay and the neutron total cross section. The loss term can also include a chemical removal term resolved in both space and time for nuclide i.

To solve the coupled system of equations given in Equation (1) in a computationally efficient manner, several simplifications are made. The results generated in this work do not account for depletion of the fuel or evolution of the neutron energy spectra [10]. Instead, these terms are assumed to be constant. Another simplification made in this work is consideration of only one spatial dimension. This includes simplifications to the flow of the fuel salt, which can only change in time and the single spatial dimension. This assumption neglects turbulent effects or vortices, which have a larger presence in some reactor designs.

2.1. Data

Two types of data are used with the method presented in this work, specifically reactor data and nuclear data. The reactor data is specific to the reactor of interest. These data include the linear flow rate, average neutron energy, average neutron flux, in-core fraction of fuel salt, ex-core fraction of fuel salt, in-core volume, ex-core volume, and average temperature. These data are used to generate the appropriate cross sections and terms in Equation (1). The solver uses the average temperature and energy to generate one-group cross sections from OpenMC.

The other source of data is general nuclear data, accessed in this work through OpenMC using the data in ENDF-B/VII.1 [10, 11]. These data include fission yield fractions, decay chains, branching ratios, half lives, and cross sections. These data are important for properly constructing the source and loss terms for nuclides, while the decay chains are useful for constructing the relationships between each nuclides.

2.2. Solver

This work uses upwind, or backwards, in space and explicit in time finite differencing to generate a coupled system of PDEs. The upwind in space approach is used to maintain numerical stability [12]. Numerical stability requires the Courant-Friedrichs-Lewy (CFL) condition shown in Equation (3) to be maintained.

$$\Delta t \leq \lambda_{CFL} \frac{\Delta z}{\nu} \tag{3}$$

Equation 3 shows that, as the spatial mesh, Δz, becomes finer or the flow rate, ν, becomes larger, the time step, Δt, must become smaller, which means higher computational cost. This is because a smaller time step becomes necessary to capture the effects spatially when the fluid moves faster through a spatial region Δz. The algorithm used to solve the coupled system of PDEs is given in Algorithm 1.

Algorithm 1 Solver Pseudocode

```
while t < t_final do
    while i ≤ i_net do
        S_i ← S_{i,new}
        μ_i ← μ_{i,new}
        N_i ← N_{i,new}
        i ← i + 1
    end while
    t ← t + Δt
end while
```

Algorithm 1 shows that, while any number of nuclides can be included in the decay chain, the computational cost will increase directly with the number of nuclides, i. Additionally, the cost scales with the time step, though the time step is directly limited by the CFL condition. The time step can be made larger by using larger spatial steps at the cost of spatial resolution.

Another point to note in the algorithm is that the source terms are updated because of the dependence on concentrations. The source terms must be updated before each subsequent concentration update, as the decay parents contribute as a source term for nuclide i. The loss rates are also updated, which is not necessary assuming constant flux and cross sections. However, for power transients, changes occur to both the fission source term and the loss term.

2.3. Scaled Flux

The scaled flux method is an approximation which can be used in the ODE, shown in Equation (2), to imitate the spatial resolution of the solve given in Equation (1) [13]. The source and loss terms are modified based on the in-core and ex-core fractions of the primary loop. The source term using the scaled flux approach, without sources from decay parents included, is given in Equation (4). This equation includes the neutron flux, ϕ; fission cross section, σ_f; independent fission yield of nuclide i, γ_i; concentration of fissile nuclide, N_f; and the in-core fraction, f_{in}.

$$S_i(t) = \phi\sigma_f\gamma_i N_f f_{in} \tag{4}$$

Equation (5) shows the loss term for the scaled flux approach. This equation includes the total cross section of nuclide i, σ_t; the decay constant, $\lambda_{d,i}$; the chemical removal rate in the ex-core region, $\lambda_{r,i}$; and the ex-core fraction, f_{out}.

$$\mu_i(t) = \phi\sigma_t + \lambda_{d,i} + \lambda_{r,i} f_{out} \tag{5}$$

3. RESULTS

For these results, the data used is shown in Table I. The fissile concentration, neutron flux, nominal power, power history, and in-core fraction come from the MCNP/ADDER simulation of the Molten Salt Reactor Experiment (MSRE) [8] The other terms come from calculations using the data from the OpenMSR project [14].

Table I. Simulation parameters used for the various simulations in this work.

Parameter	Value
Fissile Nuclide	^{235}U
Fissile Concentration	8.4×10^{19} $\frac{atoms}{cm^3}$
Neutron Flux	$8.4E12$ $\frac{n}{cm^2 s}$
Neutron Energy	0.02533 eV
Salt Density	2.32556 $\frac{g}{cm^3}$
Temperature	294 K
Nominal Power	7.34 MW
Primary Loop Flow Length	608.06 cm
In-core Fraction	27.2%
Ex-core Fraction	72.8%
Volume	2.116111 m^3
Linear Flow Rate	21.75 $\frac{cm}{s}$

3.1. Spatial Sensitivity

The spatial sensitivity analysis is useful because a sufficiently fine spatial mesh is desired for accuracy, but an excessively fine mesh will waste computational resources. To conduct this analysis, the concentration of xenon-135 after one hour in the MSRE is compared using the spatially resolved PDE method for various numbers of spatial nodes. Table II provides details on the cost and accuracy of each simulation. The computational cost scales as $O(n^2)$ with the number of spatial nodes, n. This is because a finer spatial mesh also requires a finer time stepping scheme, as shown in Equation (3).

Table II. Spatial sensitivity detailing how cost and accuracy vary based on the number of spatial nodes over one hour for xenon-135.

Nodes	Cost [s]	Inaccuracy [%]
5	0.12	20.8
10	0.29	9.11
100	6.39	0.007
200	20.4	0.002
500	111	0.001
1000	409	-

Based on the results from the spatial sensitivity study, 200 spatial nodes are used for the chain sensitivity and spatial comparison studies. For the MSRE validation, 100 spatial nodes are used instead to keep computational cost low while maintaining accuracy within 0.01%.

3.2. Chain Sensitivity

The next sensitivity study of interest is in the number of nuclides to include in the decay chain. This study is conducted by comparing the concentration of xenon-135 in the MSRE after one hour for various numbers of decay parents for xenon-135 included in the decay chain. Table III shows how varying the number of nuclides in the isobar affects the results and simulation time.

Table III. Nuclide sensitivity detailed results showing a comparison between cost and accuracy for varying numbers of nuclides in the 135 isobar over one hour.

Nuclides	Cost [s]	Inaccuracy [%]
1	4.40	84.1
2	8.30	58.4
3	13.1	31.8
4	16.9	1.08
5	20.7	0.00
6	24.5	-

The cost scales as $O(n)$ for n nuclides. Five and six nuclides have the same inaccuracy, because tin-135 does not decay into antimony-135 in the dataset used in OpenMC, so these nuclides yield the same result. From these results, using five nuclides in the 135 isobar appears to yield accurate results while using additional nuclides does not have a significant impact. For the spatial comparison, five nuclides are used for the 135 isobar leading to xenon-135.

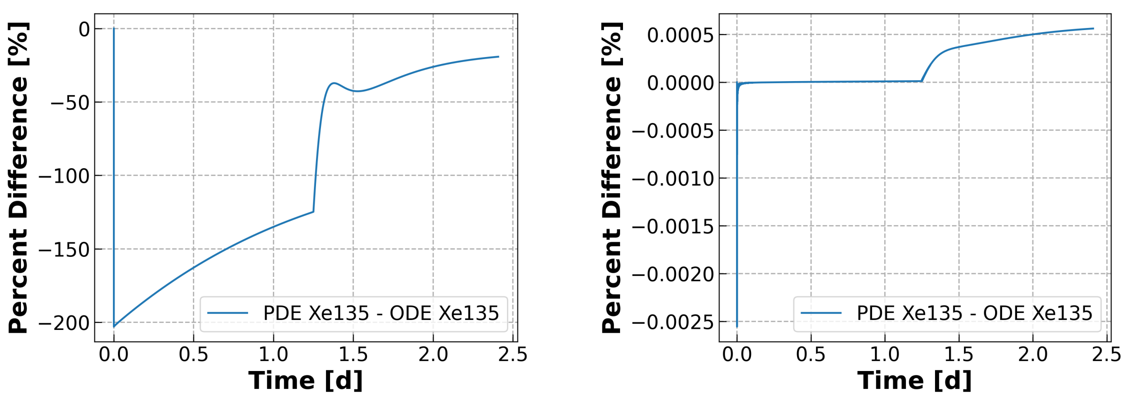
3.3. Spatial Comparison

The next results of interest are a comparison of how this method compares with a spatially unresolved method. In particular, a comparison with the scaled flux method yields insights into the difference between a spatially resolved and spatially unresolved method [13].

After 1.25 days, there is a difference of 1.2×10^{-5}% when compared to the scaled flux method and 127.7%

when compared to the ordinary differential equation method without scaling the flux. This result is consistent with the difference of 44.24% found by Tano et al. [7] which compares the unscaled flux to a quasi-static approach. The larger difference found in this work is likely due to the difference in the reactors, where Tano et al. investigate the MSFR while this work investigates the MSRE.

A step change in power is also tested to determine the impact on the spatial resolution, as the spatially resolved method may be necessary to adequately handle sudden changes in reactor conditions. After 1.25 days of depletion, the power was increased by an order of magnitude. The percent difference for the scaled flux method increased shortly after this power change, but only to approximately 0.0005%. This is shown in Figure 1, which compares the unscaled flux and scaled flux percent differences over time for the average concentration of xenon-135.



(a) Unscaled flux **(b) Scaled flux**

Figure 1. Percent difference between the spatially resolved (PDE) method and the spatially unresolved (ODE) method on the xenon-135 concentration.

From these results, it appears that using the scaled flux method is sufficient for modeling the concentrations of nuclides in MSRs even when there is a sudden change in power. However, when chemical removal is included in the ex-core region, the difference between the spatially resolved PDE and the non-spatially resolved ODE becomes non-negligible. When comparing the concentrations using both methods, there is an initial large difference which exponentially decreases to an steady-state difference between the two methods.

A further investigation into the effects of varying cycle times on the inaccuracy of the result while using the scaled flux method is provided in Table IV. The cycle time, T_{cyc}, describes the rate at which xenon is removed, where a larger cycle time corresponds to a slower removal rate. Specifically, the chemical removal rate, λ_r, is the inverse of the cycle time.

Table IV shows that, for cases where there is a greater removal rate, the scaled flux method is less accurate than a spatially refined model. This is represented both by the peak difference as well as the final difference, which occurs as the difference approaches an equilibrium value.

Table IV. The inaccuracies of using the scaled flux method to handle spatially resolved chemical removal for various chemical removal rates of xenon.

Xenon T_{cyc} $[s]$	Peak Diff [%]	Final Diff [%]
10	15.2	3.75
20	7.79	1.05
40	3.94	0.30
80	1.97	0.09

Another consideration is if there is chemical removal for the other elements in the decay chain. To simulate this, the cycle time of 20 seconds is used and then applied to the elements in the 135 isobar chain leading to xenon. Because one of those nuclides is metastable xenon-135, the case for a single nuclide in the chain having chemical removal is not modeled. The results of this study are shown in Table V.

Table V. The inaccuracies of using the scaled flux method to handle spatially resolved chemical removal for nuclides in the 135 isobar chain leading to xenon.

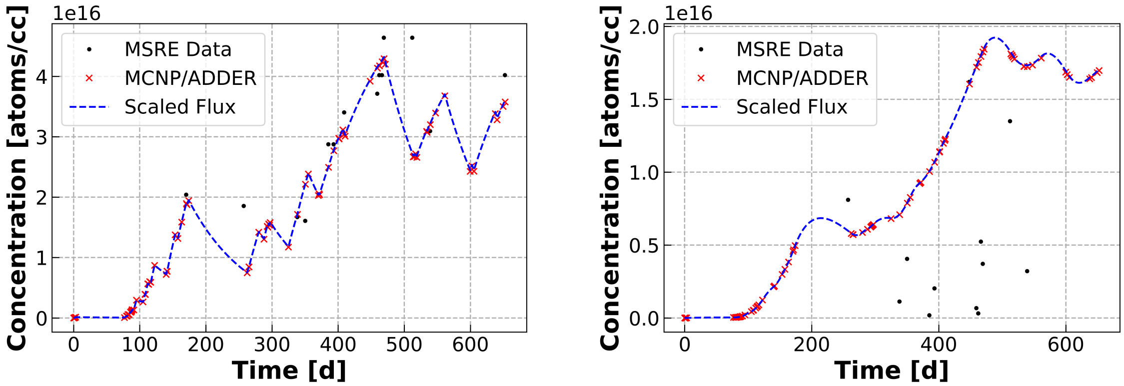
Reprocessed Nuclides	Peak Diff [%]	Final Diff [%]
0	2.8×10^{-3}	4.5×10^{-6}
2	7.8	1.1
3	7.9	2.5
4	7.9	2.6
5	7.9	2.6

3.4. Validation and Verification

Validation of this work uses the experimentally measured MSRE ladle salt samples over time during uranium-235 operation [15]. The concentrations are investigated for zirconium-95 and niobium-95 during operation in the MSRE from January 24th, 1996 to November 7th, 1967. Zirconium-95 is a salt-seeking nuclide without an appreciable noble gas precursor, which means it can be modeled without having to account for chemical removal. As discussed previously, the scaled flux method is sufficient for modeling the spatially resolved depletion without chemical removal. Not only is there a spatial component that does not need to be modeled, but the spatial resolution is also not required. This means only the decay chain sensitivity needs to be included, which has linear scaling.

For analyzing the noble metal niobium-95, chemical removal is considered in the ex-core region. The chemical removal rate used for niobium-95 in this work is the value from Shahbazi et al. of $2.3 \times 10^{-8} s^{-1}$, which accounts for the removal of elements from the fuel salt to the cover gas in the pump bowl [16]. Because this removal rate is small relative to the decay, the scaled flux method is sufficient for modeling both the zirconium-95 and niobium-95 concentrations.

The concentrations of zirconium-95 and niobium-95 over time is given in Figure 2. In the figure, comparisons are made between the experimental results of the MSRE [15], analysis from validation of NEAMS tools using MCNP/ADDER [8], and the current work using the scaled flux method.



(a) Zirconium-95 concentration

(b) Niobium-95 concentration

Figure 2. Concentrations over time in the Molten Salt Reactor Experiment.

There is good agreement between the current work and the results from MCNP/ADDER, which does not account for chemical removal. However, the MSRE experimental data does not agree at each data point. This can be explained by uncertainty in the measurements, chemical effects which aren't modeled in MCNP/ADDER or this work, and other minor contributing factors not accounted for in these models. Particularly, the niobium-95 concentrations from the MSRE measurements are smaller than the model predicts, which means that a larger chemical removal rate would likely fit these data better.

4. CONCLUSIONS

This work demonstrates a simplified approach to calculating spatially resolved nuclide concentrations in molten salt reactors, offering verification and validation of the results. The results offer close agreement with MCNP/ADDER, while the data from the MSRE does not agree as closely to either computational model. The scaled flux method was demonstrated to be effective for modeling nuclide concentrations when no spatially resolved chemical removal is included. When there is spatially resolved chemical removal, the peak percent difference scales linearly with the inverse of the cycle time. What this means practically is that most elements modeled in MSRs can be treated using the scaled flux method, while those with high chemical removal rates should be resolved spatially.

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